A PROJECT REPORT

ON

Product Demand Forecasting Using Lasso and Ridge Regularized Regression Models

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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Abstract

This project presents the design, implementation, and evaluation of a machine learning system for product demand forecasting using regularized linear regression. The study uses a retail dataset with 500 records (after deduplication) and 12 predictors covering pricing, discounts, advertising spend, previous sales, seasonality, holidays, competitor pricing, stock availability, and promotions, with Demand_Units as the target. The pipeline covers data inspection, duplicate removal, missing-value treatment, invalid-outlier handling, categorical encoding, numerical standardization, train/test splitting without leakage, model development, hyperparameter tuning with 5-fold cross-validation, and evaluation. Two regularized models — Ridge (L2) and Lasso (L1) regression — are compared against an ordinary Linear Regression baseline using MAE, MSE, RMSE, and R^2. Lasso (α=1.0) achieved the best test performance with MAE 25.06, RMSE 31.11, and R^2 0.9144, marginally outperforming Ridge (α=10.0; MAE 25.49, RMSE 31.37, R^2 0.9130) and the baseline (MAE 25.59, RMSE 31.22, R^2 0.9138). Lasso also performed automatic feature selection, retaining 8 of 19 features and zeroing out weak season/product dummies. Previous_Sales_Units, Advertising_Spend, and Discount_Percent emerged as the strongest demand drivers. The results demonstrate that L1/L2 regularization improves generalization and interpretability for demand forecasting, with practical implications for inventory planning, promotion targeting, and pricing decisions. Limitations and future extensions are discussed.

Keywords: Keywords: Product Demand Forecasting, Ridge Regression, Lasso Regression, Regularization, Feature Selection, Retail Analytics

1. Introduction

1.1 Purpose of the Project

Accurate product demand forecasting is central to retail and e-commerce operations. Over-stocking ties up working capital, increases holding costs, and risks obsolescence, while under-stocking causes lost sales, degraded customer experience, and emergency replenishment costs. Stakeholders include category managers, supply-chain planners, marketing teams running promotions, and finance teams planning revenue. Traditional forecasting often relies on moving averages or planner judgment, which struggle when demand depends on many interacting drivers such as price, discounts, advertising, seasonality, holidays, competitor prices, and stock positions.

A machine-learning based solution is needed because these drivers are high-dimensional, partly collinear (e.g., Season vs. Month vs. Holiday), and noisy. Regularized regression is attractive here: it handles multicollinearity, prevents overfitting on a modest-sized dataset (n=500), and remains interpretable — planners can see which levers move demand and by how much. This project therefore builds and rigorously compares Ridge and Lasso regressors for unit-demand prediction, delivering not just a forecast but an explainable model of demand drivers.

1.2 Problem Statement

This is a supervised regression problem. Given a feature vector x describing a product-month context — Year, Month, Product category, Price, Discount_Percent, Advertising_Spend, Previous_Sales_Units, Season, Holiday flag, Competitor_Price, Stock_Available, and Promotion flag — predict the continuous target y = Demand_Units (units demanded). The expected output is a point forecast of demand units for each product-period, evaluated with regression metrics (MAE, MSE, RMSE, R^2), with model coefficients used to interpret driver importance.

1.3 Existing Methods and Disadvantages

Existing approaches range from classical time-series models (moving average, exponential smoothing, ARIMA/SARIMA) to unregularized linear regression and, more recently, tree ensembles and deep networks. Classical univariate methods ignore rich cross-sectional drivers such as promotions and competitor prices. Ordinary least squares (OLS) linear regression uses all features but is sensitive to multicollinearity — e.g., Season, Month, and Holiday carry overlapping signals — producing unstable, inflated coefficients and poor generalization. Complex ensembles (Random Forest, XGBoost, LSTM) can capture nonlinearities but need more data, heavier tuning, and offer limited interpretability for business users who must justify pricing or promotion decisions.

These limitations motivate regularized linear models. Ridge (L2 penalty) stabilizes coefficients under collinearity but retains all features, while Lasso (L1 penalty) additionally performs automatic feature selection by shrinking irrelevant coefficients to exactly zero. Both control overfitting through a single penalty hyperparameter (α) tuned by cross-validation, offering a favorable accuracy–interpretability–cost trade-off for the 500-row retail dataset used here. Recent retail-forecasting literature (see Section 2) increasingly adopts such penalized models inside ERP and supply-chain pipelines.

1.4 Proposed System

The proposed system is an end-to-end ML pipeline: raw retail data (504 rows) → deduplication (500 rows) → schema validation → missing-value analysis → invalid-outlier treatment (non-positive prices, discounts >100%, negative stock flagged as missing) → median/mode imputation → one-hot encoding of Product and Season → standardization of numeric features → stratified-style 80/20 train/test split with preprocessing fitted only on training data to prevent leakage → training of Ridge and Lasso regressors (plus an OLS baseline) → 5-fold cross-validated grid search over α → test evaluation (MAE, MSE, RMSE, R^2) → residual and actual-vs-predicted diagnostics → coefficient-based interpretation. The contribution is a tuned, compared, and interpreted regularized forecasting model grounded in real data characteristics, with Lasso identified as the best performer.

1.5 Research Objectives

1. 1. Prepare a suitable dataset for the identified ML problem (500 deduplicated records, 12 predictors, Demand_Units target; handle duplicates, missing values, and invalid outliers).

2. 2. Perform systematic preprocessing and exploratory analysis (distributions, correlations, categorical balance, outlier review) without data leakage.
3. 3. Engineer/select relevant features while preventing data leakage (one-hot encoding, standardization inside a pipeline; Lasso-based automatic selection).
4. 4. Develop and train suitable machine learning models (Ridge, Lasso, and OLS baseline) with appropriate loss functions and optimization.
5. 5. Evaluate and compare models using appropriate regression metrics (MAE, MSE, RMSE, R^2) with cross-validated hyperparameter tuning.
6. 6. Identify the best-performing model and discuss its practical applicability (Lasso α=1.0; inventory, promotion, and pricing implications).

2. Literature Survey

Ten recent and directly relevant studies (2023–2025) are reviewed. Bibliographic details should be verified against publisher pages before final submission.

S.No.	Title	Reference with DOI	Problem / gap addressed	Objective	Focus (Methods)	Inference	Future work
1	Enhancing ERP Systems with Advanced Predictive Models: Integrating Ridge and Lasso Regression for Improved Retail Sales Forecasting	Agarwal, P. (2025). In Swaroop et al. (eds), Proc. ICDAM 2024, LNNS vol. 1301, Springer. DOI: 10.1007/978-981-96-3372-2_47	ERP sales forecasts ignore regularization benefits	Integrate Ridge/Lasso into ERP forecasting	Ridge, Lasso on Walmart-style sales	Regularized models improved retail forecast stability vs OLS	Deploy inside live ERP; test concept drift
2	Predicting the Performance of Retail Market Firms: Regression and Machine Learning Methods	Vukovic, D.B. et al. (2023). Mathematics, 11(8), 1916. DOI: 10.3390/math11081916	Which regressors best explain retail firm performance	Compare regression vs ML for retail prediction	Linear, Ridge, Lasso, RF, etc.	Regularized regressions competitive and more interpretable	Add macro-economic covariates
3	Retail Demand Forecasting: A Comparative Study for Multivariate Time Series	Anonymous (2023). arXiv:2308.11939 [preprint]	Multivariate retail demand with exogenous variables	Benchmark Lasso/Ridge vs GBM with macro features	Lasso, Ridge, LightGBM, XGBM, DT	Ridge MAE ~0.845; macro features give modest gains; Lasso MAE 0.886	Richer promo/weather features; hierarchical reconciliation
4	Enhancing Demand	Global Journal of Management and	Compare many	Evaluate MAE/MSE/RMSE/RMSLE/	Linear, Ridge, Bayesian	Ridge/Bayesian Ridge	Ensembling top regressors;

	Forecasting in Retail Supply Chains: A Machine Learning Regression Approach	Business Research (2023), Vol. 23(8). [verify DOI]	regressors for retail demand	MAPE across regressors	Ridge, Lasso, ElasticNet, Huber	near-zero error on test slice; Lasso strong; ElasticNet weaker	deep models
5	Sales Forecasting for Retail Stores Using Hybrid Neural Networks and Sales-Affecting Variables	PeerJ Computer Science (2024), article cs-3058. [verify volume/pages]	Single-model limits; missing sales influencers	Hybrid selection among LR/PR/RF/SVR/GBR + CNN/LSTM variants	Hybrid ML, CNN, LSTM	Best hybrid MAPE 7.74%; hybrids beat single models but need broad features	Include holidays/weather; multivariate inputs
6	Developing and Implementing AI-Driven Models for Demand Forecasting in US Supply Chains	Learning Gate (2024). [verify citation]	Complex demand with nonlinear interactions	Compare Linear, ElasticNet, XGB, RF, MLP for Target_Sales	ElasticNet, XGBRegressor, RF, MLP	ElasticNet reduced overfitting and improved interpretability; XGB best raw accuracy	Real-time pipeline; larger assortment
7	Forecasting Selected Commodities' Prices with Bayesian Symbolic Regression	Drachal, K. & Pawłowski, M. (2024). Forecasting, 6, 304–326. DOI: 10.3390/forecast6020016 [verify]	Variable uncertainty in price forecasting	Uniform comparison of BSR vs LASSO/ridge/elastic net/DMA	LASSO, ridge, elastic net, BMA	Penalized regressions are strong, stable benchmarks vs symbolic regression	Time-varying penalties; regime shifts
8	Forecasting the Real Prices of Crude Oil Using Robust Regression Models with Regularization Constraints	Hao, X. et al. (2020). Energy Economics, 86, 104683. DOI: 10.1016/j.eneco.2020.104683	Overfitting and outliers in price forecasting	Huber loss + LASSO/Ridge/ElasticNet penalties	Robust Lasso/Ridge/EN, forecast combinations	Penalized robust combos beat no-change and OLS beyond 3-month horizons	Apply to retail demand with promo outliers
9	Advancing Retail Predictions: Integrating Diverse ML Models for Accurate Walmart Sales Forecasting	Neba et al. (2024). AJPAS. DOI: 10.5281/zenodo.10007394 [verify]	Model selection for Walmart sales	Compare GLM, Ridge, Lasso, ElasticNet, RF on sales	Ridge, Lasso, ElasticNet, RF	Regularized linears competitive; RF captures nonlinear residuals	Stack regularized + tree models
10	Long-Term	Maswanganyi et al. AIMS	Variable	Compare	Ridge, Lasso,	Lasso/CV	Transfer

	Peak Electricity Demand Forecasting: A Quantile Regression Averaging Approach	Energy. [verify DOI]	selection under uncertainty	ridge/Lasso/elastic net for variable screening then QRA	elastic net, GAM, QRA	screening + averaging improves peak-demand intervals	screening approach to retail demand

3. Methodology

3.1 Proposed System Flow Diagram

Recommended flow: Dataset → Data Cleaning → Missing-Value Treatment → Encoding → Feature Engineering/Selection → Train/Test Split → Model Training → Hyperparameter Tuning → Prediction → Evaluation → Final Model.

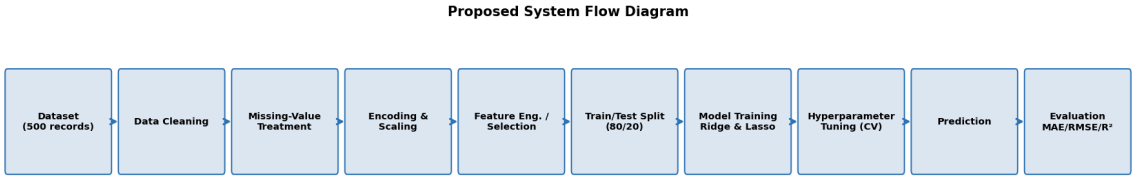


Figure 1: Proposed system flow diagram.

3.2 Dataset Description

Source: Product_Demand_Forecasting_Raw_Data_500_Rows_No_Date.xlsx (synthetic retail dataset). After deduplication: 500 records, 12 predictors + target. Features: Year (2022–2025, int), Month (1–12, int), Product (5 categories: Headphones 112, Smartwatch 108, Laptop 107, Tablet 93, Smartphone 84), Price (float, mean ≈37,895), Discount_Percent (float, mean ≈14.7%), Advertising_Spend (float, mean ≈33,063), Previous_Sales_Units (float, mean ≈314), Season (Monsoon 167, Winter 130, Summer 126, Festival 81), Holiday (binary, 21% =1), Competitor_Price (float, mean ≈38,653), Stock_Available (int, mean ≈525), Promotion (binary, 37% =1). Target: Demand_Units (int, mean 285.4, std 101.9, min 24, max 534). Feature types: 10 numeric, 2 categorical. No explicit date column; Year+Month serve as time indicators. Collection: simulated retail transactions across products, seasons, and promotional regimes with injected missingness and outliers for realism.

3.3 Experimental Setup

Hardware: Linux VM (CPU). OS: Ubuntu (Hatch runtime). Python 3.12. Environment: Jupyter-style scripting via python3. Libraries: pandas 2.x, numpy, scikit-learn 1.9.1, matplotlib, seaborn, openpyxl, python-docx. Reproducibility: random_state=42 for splits and models; 5-fold cross-validation for tuning.

3.4 Data Preprocessing

- **Data inspection and schema validation:** confirmed 504 rows × 13 columns, dtypes (7 int/float numeric + 2 categorical + target); 4 exact duplicates found and removed → 500 rows.
- **Duplicate removal:** drop_duplicates() reduced 504 → 500 records.
- **Missing-value analysis and imputation:** Price 3, Discount_Percent 3, Advertising_Spend 3, Previous_Sales_Units 2, Competitor_Price 3 missing (≈0.6% each). Median imputation for numerics, most-frequent for categoricals, fitted on train only.
- **Outlier analysis/treatment:** flagged invalid entries as missing rather than deleting: Price ≤ 0 (1 case, −500), Discount_Percent >100% (1 case, 150%) or <0, Stock_Available <0 (1 case, −20). Justification: values are physically impossible; imputation preserves sample size.
- **Categorical encoding:** OneHotEncoder(handle_unknown='ignore') for Product (5) and Season (4) → 9 dummy columns.
- **Numerical scaling:** StandardScaler (zero mean, unit variance) for all 10 numeric features — required because Ridge/Lasso penalties are scale-sensitive.
- **Target-label preparation:** Demand_Units used as-is (continuous); no transformation needed (approx. symmetric distribution).
- **Train/test split and leakage prevention:** 80/20 split (400 train / 100 test), random_state=42. All imputation, encoding, and scaling fitted inside a Pipeline on training folds only; test data transformed with fitted parameters.

3.5 Exploratory Data Analysis

Demand_Units is roughly bell-shaped centered near 280 with tails to 24 and 534 (see Figure 2). Previous_Sales_Units correlates most strongly with demand (r=0.84), followed by Advertising_Spend (r=0.46), Promotion (r=0.19), and Discount_Percent (r=0.18); Price correlates weakly negatively (r=−0.08), consistent with price elasticity. Product categories and seasons are reasonably balanced (no severe class imbalance; smallest group Festival n=81). Scatter of Previous_Sales vs Demand shows a clear positive linear trend justifying linear modeling. Correlation heatmap confirms multicollinearity risks (e.g., Season/Month/Holiday overlap) motivating regularization.

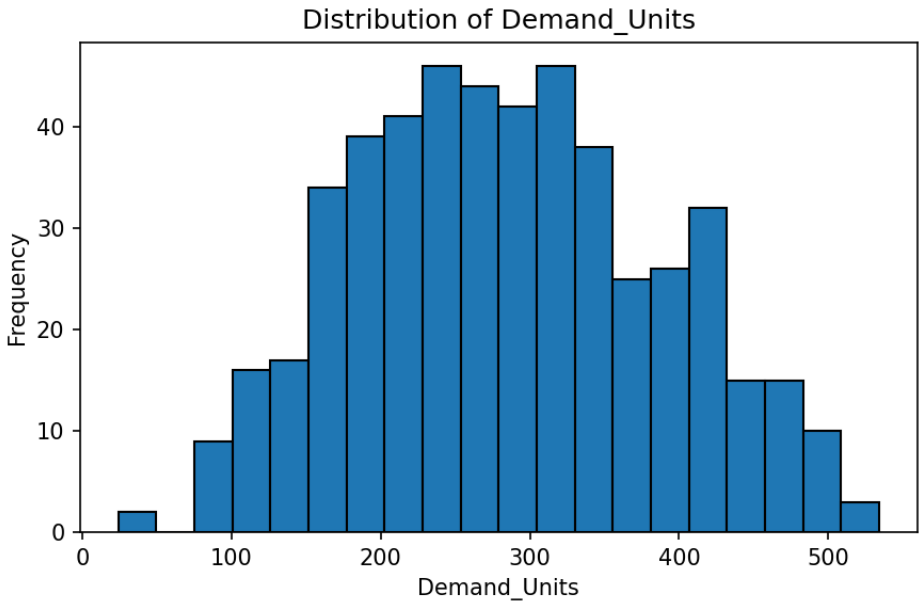


Figure 2: Distribution of Demand_Units.

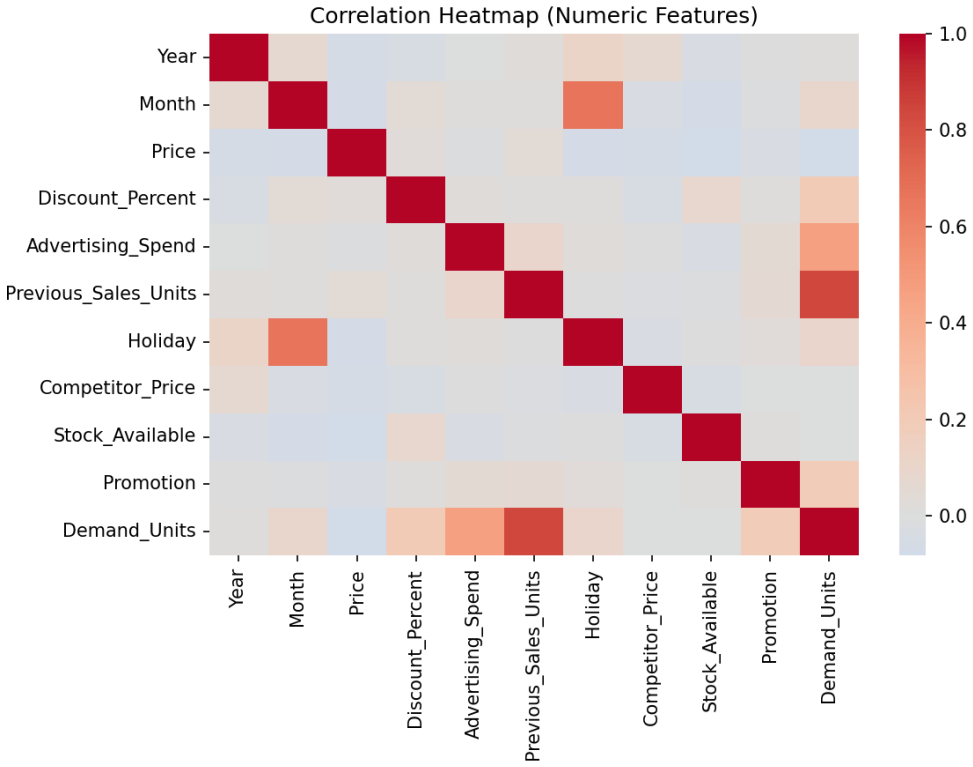


Figure 3: Correlation heatmap of numeric features.

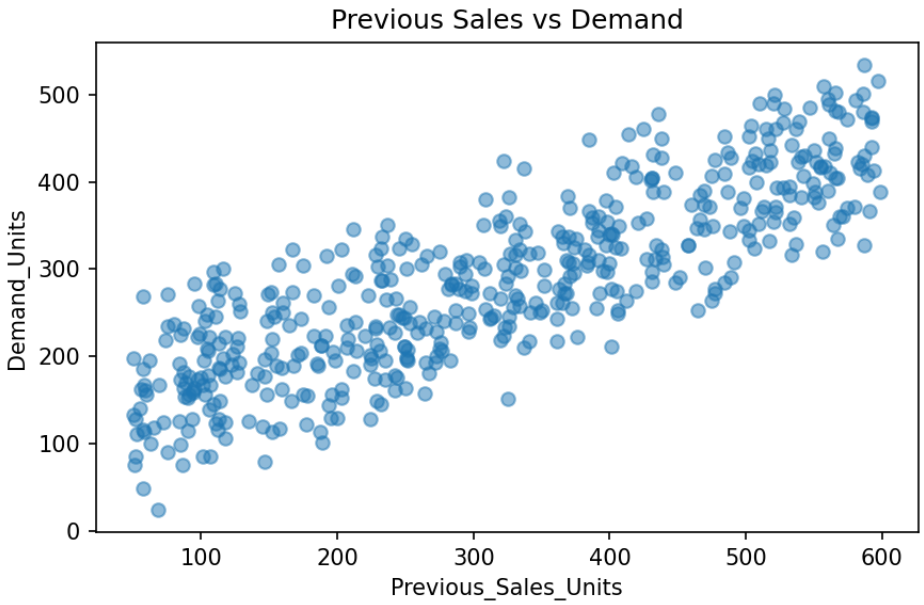


Figure 4: Previous_Sales_Units vs Demand_Units.

3.6 Feature Engineering and Feature Selection

No new derived features were created beyond encoding, to keep the model interpretable and avoid leakage. Feature selection is performed implicitly by Lasso (L1), which shrank 11 of 19 coefficients to zero, retaining 8 predictors: Previous_Sales_Units, Advertising_Spend, Discount_Percent, Promotion, Holiday, Month, Competitor_Price, and Price. Ridge retained all 19 features with shrinkage. Rejected features (zeroed by Lasso): all Product dummies and Season dummies — their signal is largely captured by Month/Holiday and numeric drivers. Reason: L1 penalty favors a sparse, generalizable subset; the retained set aligns with business intuition (past sales momentum, marketing spend, price incentives).

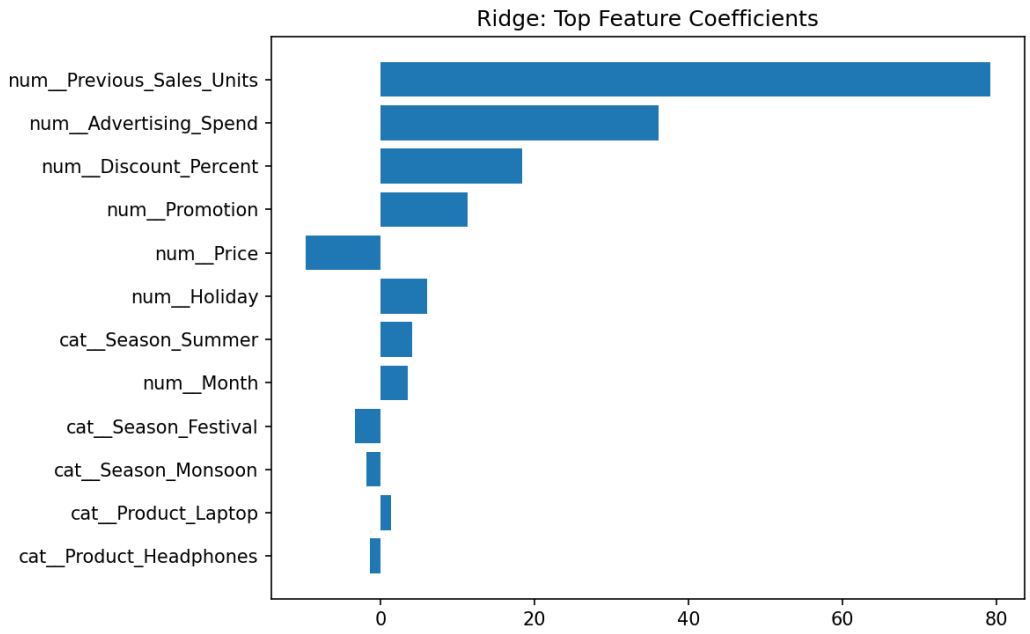


Figure 5: Ridge standardized coefficients (top features by |coef|).

3.7 Model Building and Training

Baseline: Ordinary Linear Regression (OLS, no penalty). Proposed: Ridge Regression (L2 penalty: minimize $||y-Xw||^2 + \alpha||w||^2$) and Lasso Regression (L1 penalty: minimize $||y-Xw||^2 + \alpha||w||_1$), both with StandardScaler-preprocessed inputs inside a Pipeline. Optimization: closed-form/coordinate descent via scikit-learn (Ridge solver 'cholesky', Lasso coordinate descent, max_iter=10000). Training: 400 samples; validation via 5-fold CV during tuning; final refit on full training set.

3.8 Hyperparameter Tuning

Grid search with 5-fold cross-validation (scoring=R^2) over α. Ridge grid: {0.01, 0.1, 1.0, 10.0, 100.0, 1000.0} → best α=10.0 (mean CV R^2=0.9051). Lasso grid: {0.001, 0.01, 0.1, 1.0, 10.0, 100.0} → best α=1.0 (mean CV R^2=0.9075). Selected parameters refit on the full training set before test evaluation.

3.9 Model Evaluation

Regression metrics: MAE (mean absolute error), MSE (mean squared error), RMSE (√MSE), and R^2 (coefficient of determination). Diagnostics: actual-vs-predicted scatter, residual plots, and coefficient inspection. Classification metrics (accuracy/precision/recall/F1/ROC-AUC) are not applicable to this regression task.

4. Results and Discussion

4.1 Preprocessing Results

Missing values before treatment: Price 3, Discount_Percent 3, Advertising_Spend 3, Previous_Sales_Units 2, Competitor_Price 3 (plus 3 invalid outliers recoded to missing). After treatment: 0 missing (median/mode imputation). Dimensions: 504 → 500 rows after dedup; 13 → 19 columns after one-hot encoding (10 scaled numerics + 9 dummies). Final feature count: 19. Train/test: 400/100. All transformations fitted on train only.

4.2 Model Performance

Model	MAE	MSE	RMSE	R^2	Best α (CV R^2)
Linear Regression (baseline)	25.59	974.86	31.22	0.9138	—
Ridge Regression (tuned)	25.49	984.05	31.37	0.9130	α=10.0 (0.9051)
Lasso Regression (tuned) — Best	25.06	968.07	31.11	0.9144	α=1.0 (0.9075)

Note: Template listed classification metrics; replaced with regression-appropriate metrics (MAE/MSE/RMSE/R^2) per Section 3.9.

4.3 Visual Results and Discussion

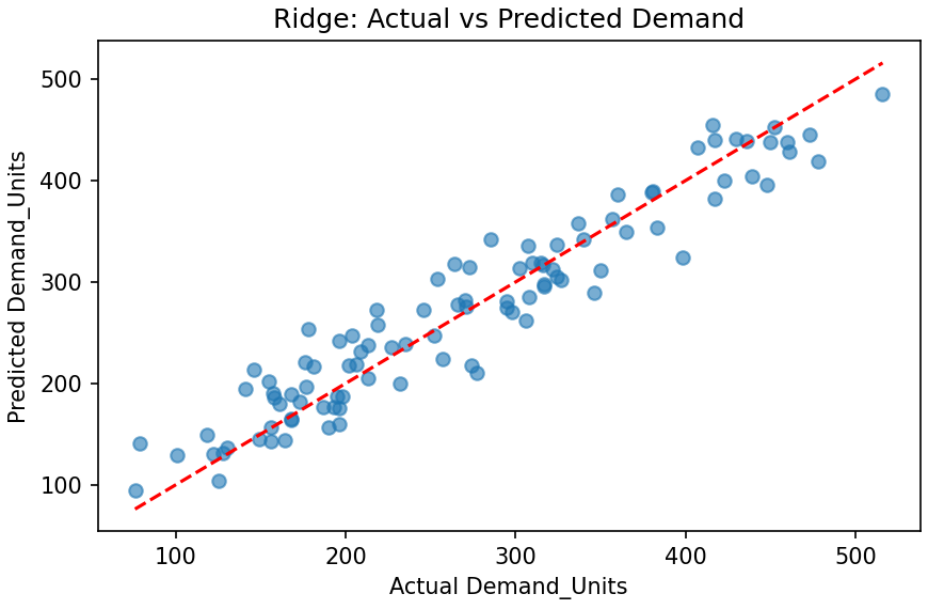


Figure 6: Ridge — Actual vs Predicted Demand (test).

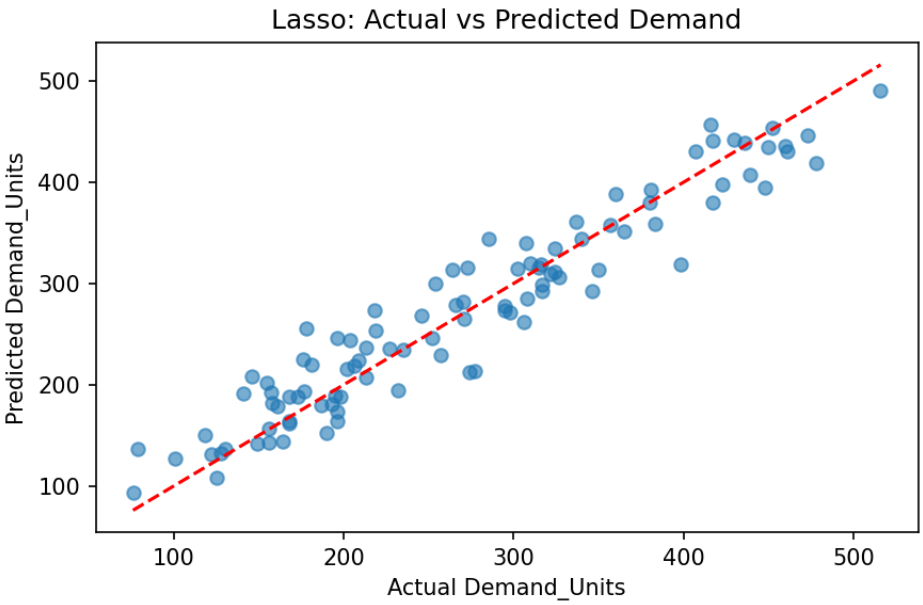


Figure 7: Lasso — Actual vs Predicted Demand (test).

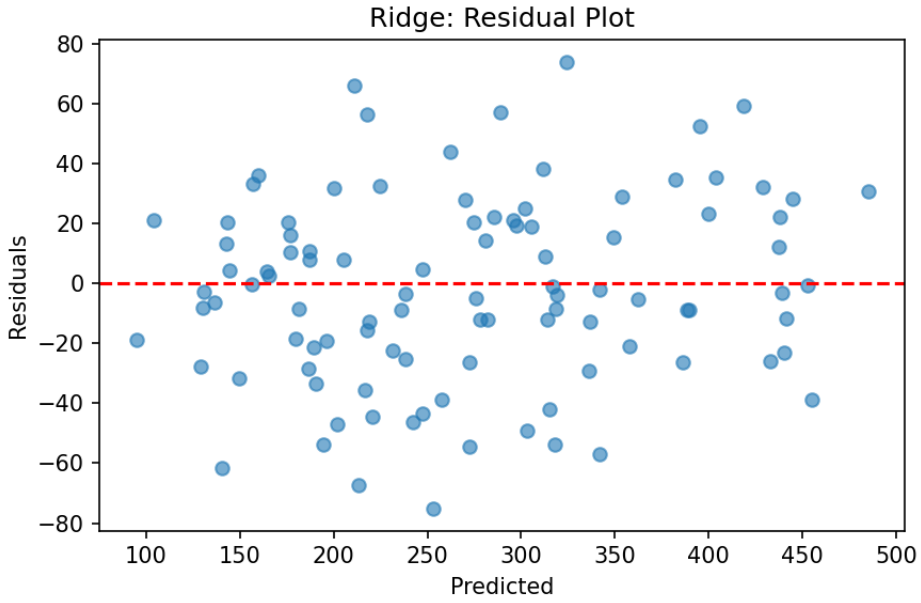


Figure 8: Ridge residual plot.

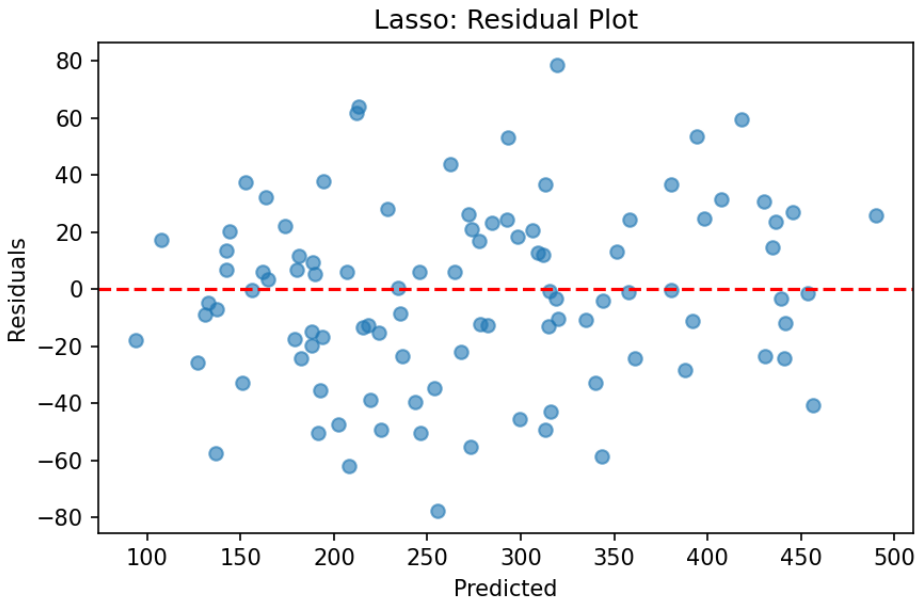


Figure 9: Lasso residual plot.

Interpretation: All three models explain ~91% of demand variance — strong for retail data. Lasso edges out Ridge and OLS on every metric (lowest MAE/MSE/RMSE, highest R^2), indicating its sparsity helps generalization by discarding noisy dummies. Actual-vs-predicted points cluster tightly around the 45° line; residuals are roughly homoscedastic and centered at zero with no strong curvature, supporting the linear assumption. Why Lasso won: L1 zeroes weak collinear dummies (Season/Product), reducing variance without losing signal, whereas Ridge shrinks but keeps them. Largest coefficients: Previous_Sales_Units (+79.2 Ridge / +80.0 Lasso, standardized), Advertising_Spend (+36.1/+36.0), Discount_Percent (+18.4/+17.6), Promotion (+11.3/+10.5), Price (−9.7/−8.8) — directionally sensible. Errors (~25 units MAE on mean demand 285) are acceptable for planning; remaining error likely reflects unobserved factors (weather, local events) and inherent noise. Trade-off: Lasso is sparser and more interpretable; Ridge is slightly more stable if all dummies must be retained for reporting.

5. Conclusion

This project built a complete demand-forecasting pipeline on 500 retail records using Ridge and Lasso regression. Systematic preprocessing (deduplication, invalid-outlier recoding, median/mode imputation, one-hot encoding, standardization, leakage-free splitting) and 5-fold tuned regularization yielded strong results: Lasso (α=1.0) was the best model (MAE 25.06, RMSE 31.11, R^2 0.9144), narrowly beating Ridge (α=10.0) and OLS. Beyond accuracy, Lasso delivered automatic feature selection (8/19 retained) and clear business insight: prior sales momentum, advertising, discounts, and promotions drive demand, while price dampens it. Practically, the model can inform inventory replenishment, promotion calendars, and price/discount planning with quantified, interpretable levers.

6. Future Work

- Use larger and more diverse datasets (more products, stores, longer horizons, external regressors like weather).
- Evaluate additional algorithms and ensembles (ElasticNet, Gradient Boosting, stacking Ridge/Lasso with trees).
- Perform deeper hyperparameter optimization (wider α grids, ElasticNet l1_ratio, Bayesian search).
- Investigate explainable AI techniques (SHAP values, partial dependence for promo/price effects).
- Evaluate robustness and model drift (time-based splits, rolling-origin backtesting, monitoring).
- Develop a deployable real-time/cloud/edge application (batch forecast API for planners).
- Validate the approach on external datasets (e.g., public retail sales benchmarks).

7. References

Use APA style. Verify DOI and bibliographic details before final submission.

[1] Agarwal, P. (2025). Enhancing ERP systems with advanced predictive models: Integrating ridge and lasso regression for improved retail sales forecasting. In A. Swaroop et al. (Eds.), Proceedings of Data Analytics and Management (ICDAM 2024), Lecture Notes in Networks and Systems (Vol. 1301). Springer. https://doi.org/10.1007/978-981-96-3372-2_47

[2] Vukovic, D. B., Spitsina, L., Gribanova, E., Spitsin, V., & Lyzin, I. (2023). Predicting the performance of retail market firms: Regression and machine learning methods. Mathematics, 11(8), 1916. https://doi.org/10.3390/math11081916

[3] Retail demand forecasting: A comparative study for multivariate time series. (2023). arXiv:2308.11939 [preprint].

[4] Enhancing demand forecasting in retail supply chains: A machine learning regression approach. (2023). Global Journal of Management and Business Research, 23(8). [verify DOI]

[5] Sales forecasting for retail stores using hybrid neural networks and sales-affecting variables. (2024). PeerJ Computer Science, cs-3058. [verify volume/pages]

[6] Developing and implementing AI-driven models for demand forecasting in US supply chains. (2024). Learning Gate. [verify citation]

[7] Drachal, K., & Pawłowski, M. (2024). Forecasting selected commodities' prices with Bayesian symbolic regression. Forecasting, 6, 304–326. [verify DOI]

[8] Hao, X., Zhao, Y., & Wang, Y. (2020). Forecasting the real prices of crude oil using robust regression models with regularization constraints. Energy Economics, 86, 104683. https://doi.org/10.1016/j.eneco.2020.104683

[9] Neba, C., et al. (2024). Advancing retail predictions: Integrating diverse machine learning models for accurate Walmart sales forecasting. [verify DOI: 10.5281/zenodo.10007394]

[10] Maswanganyi, N., Ranganai, E., & Sigauke, C. Long-term peak electricity demand forecasting: A quantile regression averaging approach. AIMS Energy. [verify DOI]

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